

Project Proposal : CS 7643
Deep Learning for Uplift Modeling: Predicting Incremental Customer Response to Marketing Interventions
Team: Neural Noobs

Joshua Harasaki
joshua.harasaki@gatech.edu

Srishti Jain
sjain443@gatech.edu

Jorge Moya
jverdugo6@gatech.edu

Karan Swamy
kswamy6@gatech.edu

Abstract

Marketing organizations invest significant resources in promotions, loyalty programs, retention campaigns, and personalized offers. Traditional machine learning models are commonly used to predict customer outcomes such as spending, engagement, or churn. However, these models do not answer the key business question:

"Which customers will change their behavior because of the intervention?"

This project aims to estimate customer-level treatment effects, also known as uplift. Rather than predicting who is likely to spend money, we seek to predict which customers will generate incremental value when exposed to a marketing action.

This problem is particularly important because marketing budgets are limited. Sending promotions to customers who would have behaved the same way without treatment creates unnecessary costs and reduces campaign efficiency.

methodologies have important limitations. Pre-versus-post comparisons are highly susceptible to confounding factors, making it difficult to isolate the true effect of the incentive. Similarly, predictive machine learning models are designed to identify patterns associated with an outcome, but they do not establish causality. In practice, they answer questions such as:

"Who is likely to spend money?"

rather than:

"Who spent money because of the incentive?"

As a result, these approaches struggle to distinguish between organic behavior and behavior that was genuinely influenced by a marketing intervention.

The standard solution to this problem is randomized controlled experimentation (A/B testing), where outcomes are compared between treatment and control groups. While A/B testing provides a reliable estimate of the average treatment effect, it is typically analyzed at an aggregate level. This allows us to determine whether an incentive was effective overall, but it does not explain which individual customers were influenced or quantify the treatment effect for each customer.

Recent advances in causal machine learning and deep learning seek to address this limitation. Architectures such as TARNet and DragonNet are specifically designed to estimate individualized treatment effects, enabling organizations to identify not only whether an intervention works, but also which customers are most likely to change their behavior because of it.

This project investigates whether deep learning approaches can successfully estimate these individual-

1. Project Summary/Background/Motivation

I. What problem did you try to solve?

Our goal is to move beyond traditional predictive analytics and address a fundamental question in marketing and customer analytics:

Is the incentive causing the behavior?

While the question appears simple, answering it correctly is significantly more challenging than it may seem.

Traditional approaches often rely on pre-versus-post analyses or predictive machine learning models to estimate the impact of an intervention. However, these

level causal effects and outperform traditional predictive and uplift modeling techniques.

II. Why is it interesting?

This problem is interesting because it addresses one of the most important questions behind any initiative designed to influence customer behavior: *what is the actual impact of the intervention?*

The ability to distinguish between organic behavior and behavior caused by an incentive has significant implications for strategy, product development, and marketing. Organizations regularly invest substantial resources in promotions, loyalty programs, retention campaigns, and product improvements, yet they often struggle to determine whether observed outcomes were caused by these initiatives or would have occurred naturally. Accurately measuring causal impact enables data scientists and decision-makers to focus on interventions that truly drive customer behavior and business value.

From a business perspective, causal inference represents a major advancement over traditional analytics. Rather than simply measuring outcomes, it seeks to understand the mechanisms that generate those outcomes. This allows organizations to allocate resources more effectively, personalize interventions, and design more impactful customer experiences.

From a technical perspective, this project provides an opportunity to explore a domain where deep learning may offer unique advantages. Most business analytics problems are traditionally dominated by classical machine learning methods because the data is primarily tabular, explainability is important, datasets are often moderate in size, and solutions must be maintainable by business analytics teams.

For common business applications such as customer segmentation, churn prediction, response modeling, customer lifetime value estimation, marketing mix modeling, and attribution, practitioners typically begin with methods such as Logistic Regression, Random Forests, or gradient-boosted trees such as XGBoost, LightGBM, and CatBoost. These models are often highly effective and difficult to outperform.

However, while these approaches excel at prediction, they are not designed to estimate individualized causal effects. They can identify customers who are likely to spend money, respond to a campaign, or churn, but they cannot determine whether a specific intervention caused that behavior. Estimating individual treatment effects requires modeling counterfactual outcomes, a fundamentally different and more challenging problem.

III. Why is it appropriate for Deep Learning?

Recent advances in deep learning, including architectures such as TARNet [1] and DragonNet [2], have been specifically developed to address this challenge. These models attempt to learn representations that support individualized treatment effect estimation, making them particularly relevant for causal inference applications.

Unlike traditional predictive models, these architectures are explicitly designed to estimate counterfactual outcomes and heterogeneous treatment effects at the individual level. By learning rich latent representations of customers, they can capture complex non-linear interactions between customer characteristics, treatment assignment, and outcomes that may be difficult to model using classical approaches.

As a result, this project offers a compelling opportunity to apply deep learning to a business problem where traditional predictive models may not be sufficient and where the potential value of accurate causal estimation is exceptionally high.

2. Approach

We plan to compare several established causal inference and uplift modeling approaches with a custom deep learning implementation developed in PyTorch.

The baseline methods will include traditional treatment-effect estimation techniques such as Causal Forests [3], as well as state-of-the-art deep learning architectures including TARNet and DragonNet. These models have been specifically designed to estimate individualized treatment effects and serve as strong benchmarks for our evaluation.

In addition to these approaches, we will develop our own neural network architecture using PyTorch. Rather than relying exclusively on existing libraries, implementing the model ourselves will allow us to gain a deeper understanding of the underlying mechanisms used to estimate counterfactual outcomes and treatment effects. This approach will also provide greater transparency into the modeling process, enabling us to better analyze how representations are learned and how treatment effects are generated.

Our primary objective is not necessarily to outperform existing state-of-the-art methods, but rather to investigate whether a custom implementation can achieve competitive performance while improving interpretability and understanding of the treatment-effect estimation process. Since causal inference models are often used to support high-impact business decisions, transparency and explainability are important considerations alongside predictive performance.

3. Resources / Related Work and Papers

I. TARNet

Uri Shalit, Fredrik D. Johansson, David Sontag. "Estimating individual treatment effect: generalization bounds and algorithms."

II. DragonNet

Claudia Shi, David M. Blei, Victor Veitch. "Adapting Neural Networks for the Estimation of Treatment Effects."

III. Uplift Modeling

Radcliffe, Nicholas. "Using Control Groups to Target on Predicted Lift: Building and assessing uplift model"

4. Dataset

I. Option 1: Kevin Hillstrom MineThatData E-MailAnalytics

This dataset [4] contains 64,000 customers who last purchased within twelve months. The customers were involved in an e-mail test.

1/3 were randomly chosen to receive an e-mail campaign featuring Mens merchandise.

1/3 were randomly chosen to receive an e-mail campaign featuring Womens merchandise.

1/3 were randomly chosen to not receive an e-mail campaign.

During a period of two weeks following the e-mail campaign, results were tracked. The task is to tell the world if the Mens or Womens e-mail campaign was successful.

II. Option 2: Criteo Uplift Modeling Dataset

This dataset [5] is constructed by assembling data resulting from several incrementality tests, a particular randomized trial procedure where a random part of the population is prevented from being targeted by advertising. it consists of 25M rows, each one representing a user with 11 features, a treatment indicator and 2 labels (visits and conversions).

III. Option 3: Starbucks promotional campaign dataset

The Starbucks promotional campaign dataset [6] is a simulated collection of data, popularized by the Udacity Capstone Challenge and Kaggle, mimicking customer behavior on the Starbucks Rewards mobile app over a 30-day period

5. Evaluation Plan

Evaluating individualized treatment effect models is challenging because true individual treatment effects are fundamentally unobservable. For each customer, we only

observe the outcome under either treatment or control, but never both simultaneously. As a result, standard classification metrics such as accuracy, precision, recall, and confusion matrices are insufficient for evaluating treatment-effect estimation directly.

Instead, we will evaluate the models using established uplift modeling and causal inference metrics, including:

- Qini Curves
- Qini Coefficient
- Uplift Curves
- AUUC (Area Under the Uplift Curve)

These metrics measure how effectively a model identifies customers who are most likely to exhibit incremental behavior due to treatment.

In addition, we will perform a business-oriented evaluation by ranking customers according to their predicted treatment effect and simulating targeted marketing campaigns. We will compare the estimated incremental outcomes achieved by targeting the top 5%, 10%, and 20% of customers identified by each model.

As a secondary evaluation, we will assess the predictive performance of the outcome models used within each causal framework. For these auxiliary prediction tasks, standard classification metrics such as AUC, precision, recall, and accuracy may be reported; however, they will not serve as the primary measure of treatment-effect estimation quality.

5.1. Risks

The primary risks associated with this project include:

- **Difficulty reproducing published architectures.** Models such as TARNet and DragonNet involve specialized training procedures and architectural choices that may be challenging to reproduce faithfully within the project timeline.
- **Evaluation complexity due to unobservable counterfactuals.** Individual treatment effects cannot be directly observed, making evaluation significantly more difficult than traditional supervised learning problems. The project will rely on uplift-specific metrics and indirect evaluation methodologies.
- **Limited performance gains over simpler approaches.** It is possible that deep learning models may not significantly outperform traditional causal inference methods such as Causal Forests. Given that many business datasets are tabular and moderate in size, simpler methods may remain competitive.

- **Negative research outcome.** A possible outcome of this project is finding that deep learning approaches do not provide meaningful improvements over traditional causal inference techniques for this application. While such a result would not validate the original hypothesis, it would still provide valuable insights regarding the practical applicability of deep learning for treatment-effect estimation in business settings.

6. Ethical Implications

Uplift modeling may influence how organizations allocate marketing resources, incentives, and customer interactions. While these models have the potential to improve the effectiveness of marketing interventions, they also introduce several ethical considerations.

Potential concerns include:

- **Unequal treatment of customer groups.** If historical data contains biases or systematic differences across customer populations, the model may disproportionately allocate incentives to certain groups while excluding others.
- **Lack of transparency in personalized decision-making.** Deep learning models can be difficult to interpret, making it challenging for stakeholders to understand why certain customers are targeted while others are not. This lack of transparency may reduce trust and accountability.

To mitigate these risks, we will focus on methodological evaluation rather than deployment recommendations and emphasize model transparency and interpretability throughout the analysis. We will also discuss the limitations of the proposed approaches and the potential consequences of applying such models in real-world decision-making environments.

References

- [1] U. Shalit, F. D. Johansson, and D. Sontag, “Estimating individual treatment effect: Generalization bounds and algorithms,” in *Proceedings of the 34th International Conference on Machine Learning*, pp. 3076–3085, PMLR, 2017. 2
- [2] C. Shi, D. M. Blei, and V. Veitch, “Adapting neural networks for the estimation of treatment effects,” in *Advances in Neural Information Processing Systems*, 2019. 2
- [3] S. Wager and S. Athey, “Estimation and inference of heterogeneous treatment effects using random forests,” *Journal of the American Statistical Association*, vol. 113, no. 523, pp. 1228–1242, 2018. 2
- [4] K. Hillstrom, “Minethatdata e-mail analytics and data mining challenge.” <https://blog.minethatdata.com/2008/03/minethatdata-e-mail-analytics-and-data.html>, 2008. 3
- [5] Diemert Eustache, Betlei Artem, C. Renaudin, and A. Massih-Reza, “A large scale benchmark for uplift modeling,” in *Proceedings of the AdKDD and TargetAd Workshop, KDD, London, United Kingdom, August, 20, 2018*, ACM, 2018. 3
- [6] Starbucks and Udacity, “Starbucks offers advertisement data.” <https://www.kaggle.com/datasets/mexwell/starbucks-offers-advertisement-data>, 2018. Simulated data mimicking customer behavior on the Starbucks rewards mobile app, provided for the Udacity Capstone Challenge. 3